Prototyping Area

Draw your project schematic here

Headers / Solder Pads

Power
- 1
- 2 5V IOREF
- 3 RESET
- 4 3.3V
- 5 5V
- 6 GND
- 7 GND
- 8 VIN

H2

3V3

Analog
- 1 A0
- 2 A1
- 3 A2
- 4 A3
- 5 A4_SDA
- 6 A5_SCL

H3

D10, 11, 12, 13 used by USB Host interface.
D9 used by USB Host I/C INT signal.
D4 used by microSD card Chip Select.

C1 100nF 0603 GND

M100.1
- 10 A5_SCL
- 9 A4_SDA
- 8 AREF
- 7 GND
- 6 D13_SCK
- 5 D12_MISO
- 4 D11_MOSI
- 3 D10_SS
- 2 D9 USB_INT
- 1 D8

H1

Digital_0_7
- 8 D7
- 7 D6
- 6 D5
- 5 D4 uSD_nCS
- 4 D3
- 3 D2
- 2 D1_TX
- 1 D0_RX

H4

Headers / Solder Pads

3V3 Power Supply 800mA

REG1 LD1117-SOT89 (IN, OUT, ADJ, TAB)
Dual Footprint
REG2 LD1117-SOT223 (IN, OUT, ADJ, TAB)
5V → 3V3
C7 1uF 0603
C8 10uF 6.3V A TANT (3.3V)
GND

USB Host Port for Android Device or USB Device

VBUS/5V/9V
R5 1K 0603
VBUS
GND
USB (SHELL2, SHELL1, D-, D+, GND)
R6 33R 1% 0603
R7 33R 1% 0603
0805 INDUCTOR
C15 18pF 0603
C16 18pF 0603
X2 12MHz
R8 2K2 0603
3V3

Connect for USB Power Out
JP1 VBUS
C12 100nF 0603
C13 10uF 6.3V A TANT
500mA PTC 1812 F2
>500mA Ferrite Inductor 0805
5V

C19 100nF 0603
5V-> (VCC, 1A, 2A, 3A, 4A, 5A, 6A)
3V3-> (GND, 1Y, 2Y, 3Y, 4Y, 5Y, 6Y)
- USB_RESET
- D11_MOSI
- D13_SCK
- D10_SS
- RES_3V3
- MOSI_3V3
- SCK_3V3
- SS_3V3

IC4 MAX3421ESMD
MAX3421 @ 3.3V VCC
VOH min 2.90V
VOL max 0.40V
ATmega328 VIH typ @ 5V VCC >= 2.65V
ATmega328 VIL typ @ 5V VCC <= 2.10V

C20 100nF 0603
C21 100nF 0603

H5
- GPIN0 26 / 1
- GPIN1 27 / 2
- GPIN2 28 / 3
- GPIN3 29 / 4
- GPIN4 30 / 5
- GPIN5 31 / 6
- GPIN6 32 / 7
- GPIN7 1 / 8

H6
- GOUT0 4 / 1
- GOUT1 5 / 2
- GOUT2 6 / 3
- GOUT3 7 / 4
- GOUT4 8 / 5
- GOUT5 9 / 6
- GOUT6 10 / 7
- GOUT7 11 / 8

USB_INT 18 / RES_3V3 12 / USB_GPX 17 / MOSI_3V3 16 / D12_MISO 15 / SCK_3V3 13 / SS_3V3 14
VCC 2 / VL 3 / GND@3 19 / GND@19 / VBCOMP 22

<-- USB_RESET → SJ4 (RESET) Default: RESET
--> USB_INT → SJ5 (D7) Default: D9
--> USB_GPX → USB_GPX0 M01PTH (D3)

Freetronics USBDroid R3
www.freetronics.com/usbdroid

DC Regulated Power Supply

D1 1N4004 SMB
C3 47uF 25V
5.0V 1.5A LDO IC1 LD29150DT50R (IN, OUT, GND)
C5 100nF 0603
C4 47uF 25V
LED1 PWR/BLUE
RN2A 1K

DCInput/USBVCC Auto-switch

> 5.4V @ VIN = DCInput selected
RN1C 10K
R11 18K 0603
RN1D 10K
Q1 MOSFET-PCHANNEL
C11 100nF 0603

In Circuit Serial Programming

P2 ICSP:MAIN (D12_MISO, D13_SCK, RESET; 5V, D11_MOSI, GND)
P4 ICSP:USB (MISO2, SCK2, RESET2; 5V, MOSI2, GND)

micro SD Card Slot

USD-SUCKETNEW (NC, CS, DI, VCC, SCK, GND, DO, RSV, CD1, CD2, SHIELD1, SHIELD2, GND1, GND2)
RN2D 1K D4 uSD_nCS
RN2C 1K D11_MOSI
RN2B 1K D13_SCK
RN2A 1K D12_MISO
RN4B 2K2, RN4C 2K2, RN4A 2K2
3V3
C23 100nF 0603
CD0 M01PTH

ATmega16U2 Microcontroller

RN1A 10K
RESET2
R1 1M 0603
C2 22pF 0603
X1 16MHz
C6 22pF 0603
R2 27R 0603
C9 1uF 0603
C10 100nF 0603
500mA PTC 1812 F1
USB-6PSMT
PGB1010603 Z1, Z2
22R 0603 R3, R4
USBVCC, USBGND
(Linked) SJ1 GROUND
0805 INDUCTOR
ATMEL_ATMEGABU2-MUQFN

PB7 21 / PB6 20 / PB5 19 / PB4 18 / (MISO)PB3 17 MISO2 / (MOSI)PB2 16 MOSI2 / (SCLK)PB1 15 SCK2 / (SS)PB0 14
PC7 22 / PC6 23 / PC5 25 / PC4 26 / PC2 5
(CTS)PD7 13 / (RTS)PD6 12 / (XCK)PD5 11 TXLED / PD4 10 RXLED / (TXD1)PD3 9 USBRXD / (RXD1)PD2 8 USBTXD / PD1 7 / PD0 6

LED2 TX/YEL RN3A 1K
LED3 RX/GRN RN3B 1K

ATmega328P Microcontroller

IC5
Short across SJ3 RESET-EN and SJ1 GROUND after all other routing is complete
SJ2 UBOOT (Linked) SJ3 RESET-EN
S1 RESET
RN1B 10K
C14 100nF 0603
C17 22pF 0603
C18 22pF 0603
X3 16MHz
R9 1M 0603
C22 100nF 0603
RN4D 2K2

PC6(/RESET) 29 / AGND 21 / AREF 20 / AVCC 18 / PB6(XTAL1/TOSC1) 7 / PB7(XTAL2/TOSC2) 8 / GND@1 3 / GND@2 5 / VCC@1 4 / VCC@2 6

PC0(ADC0) 23 A0 / PC1(ADC1) 24 A1 / PC2(ADC2) 25 A2 / PC3(ADC3) 26 A3 / PC4(ADC4/SDA) 27 A4_SDA / PC5(ADC5/SCL) 28 A5_SCL / ADC6 19 / ADC7 22
PD0(RXD) 30 D0_RX / PD1(TXD) 31 D1_TX / PD2(INT0) 32 D2 / PD3(INT1) 1 D3 / PD4(XCK/T0) 2 D4 uSD_nCS / PD5(T1) 9 D5 / PD6(AIN0) 10 D6 / PD7(AIN1) 11 D7
PB0(ICP) 12 D8 / PB1(OC1A) 13 D9 USB_INT / PB2(SS/OC1B) 14 D10_SS / PB3(MOSI/OC2) 15 D11_MOSI / PB4(MISO) 16 D12_MISO / PB5(SCK) 17 D13_SCK

AVR-MEGA8-AI

RN3C 1K, RN3D 1K
M01PTH A1, M01PTH A2
D13 LED
R10 1K 0603
Q2 MOSFET-NCHANNELSMD

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